

JKBOSE 2026

CLASS 11TH

THE LIVING WORLD

LECTURE- 05

BOTANY

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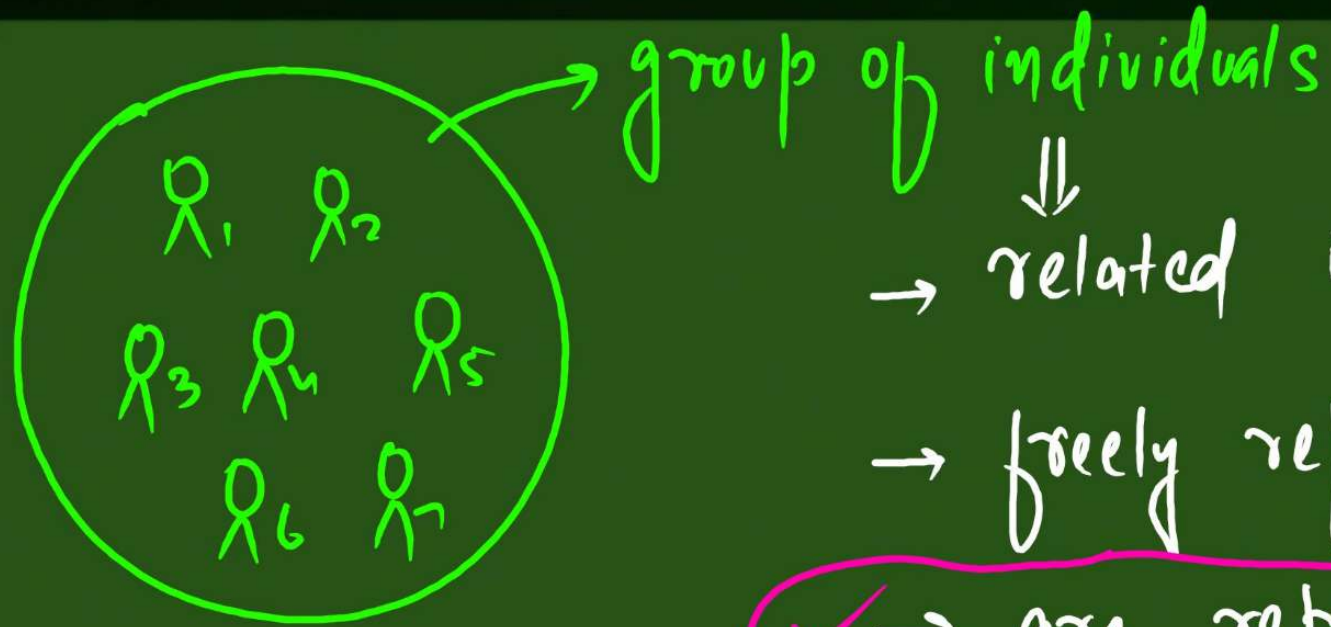
Topics To Covered

1 Taxonomic Categories (Species, Genus, Family, Order, Class, Phylum/Division, Kingdom)

2 Herbarium

→ +nt in JK BOARD Syllabus

→ Not in NEET Syllabus.



→ related in terms of Str. and functioning.

→ freely reproduce & produce fertile offsprings.

✓ → are reproductively isolated from other sps.

⇓
Main basis of biological
Species Concept.

SPECIES



÷ Real basic unit of classification.

* NOTE: Unit of classification — TAXON.

÷ Species — group of individual organisms with fundamental similarities as a species.

↓
defined by "Biological Species Concept" — proposed by "Ernst Mayr"

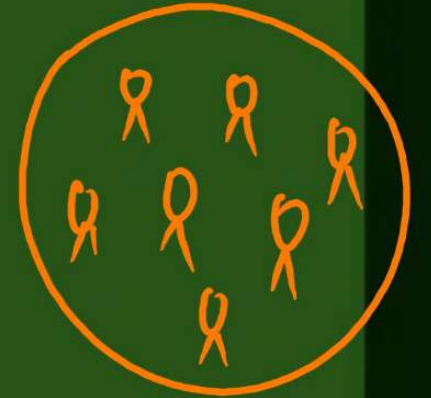
⇓
As per this concept, species is a group of organisms which:

are

a. Closely related (structurally and functionally)

b. Can interbreed freely in nature & produce fertile offsprings

c. are reproductively isolated from other sps. (Main basis of biological sps. concept)



1.2.1 Species

Taxonomic studies consider a group of individual organisms with fundamental similarities as a **species**. One should be able to distinguish one species from the other closely related species based on the distinct morphological differences. Let us consider *Mangifera indica*, *Solanum tuberosum* (potato) and *Panthera leo* (lion). All the three names, *indica*, *tuberosum* and *leo*, represent the specific epithets, while the first words *Mangifera*, *Solanum* and *Panthera* are genera and represents another higher level of taxon or category. Each genus may have one or more than one specific epithets representing different organisms, but having morphological similarities. For example, *Panthera* has another specific epithet called *tigris* and *Solanum* includes species like *nigrum* and *melongena*. Human beings belong to the species *sapiens* which is grouped in the genus *Homo*. The scientific name thus, for human being, is written as *Homo sapiens*.

NOTE :

1. Biological Species Concept is not applicable everywhere.

Ex. Mule = ♂ Donkey x ♀ horse
(Sterile)

2. Ernst Mayr
(1904 - 2004)

→ Kld as "Darwin of 20th Century"

→ Evolutionary Biologist.

→ Awarded by 03 prizes (AKA Triple Crown of biology)

a. Balzan Prize (1983) ✓

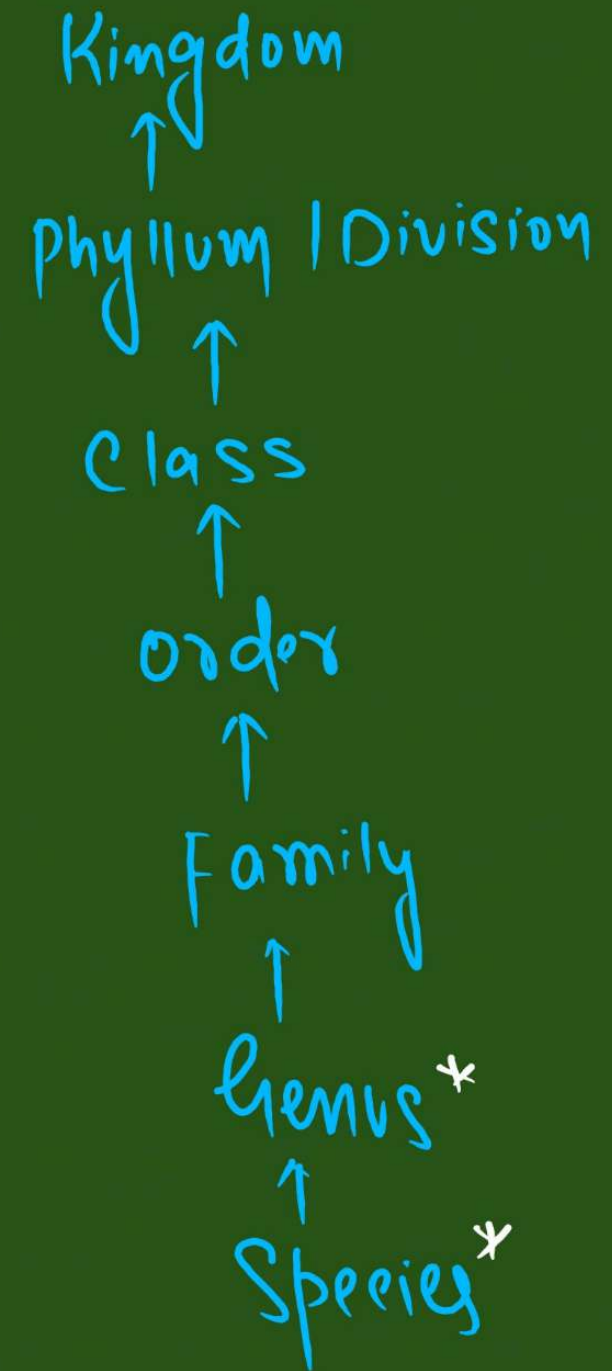
b. International Prize for Biology (1994) ✓

c. Crafoord Prize (1999) ✓



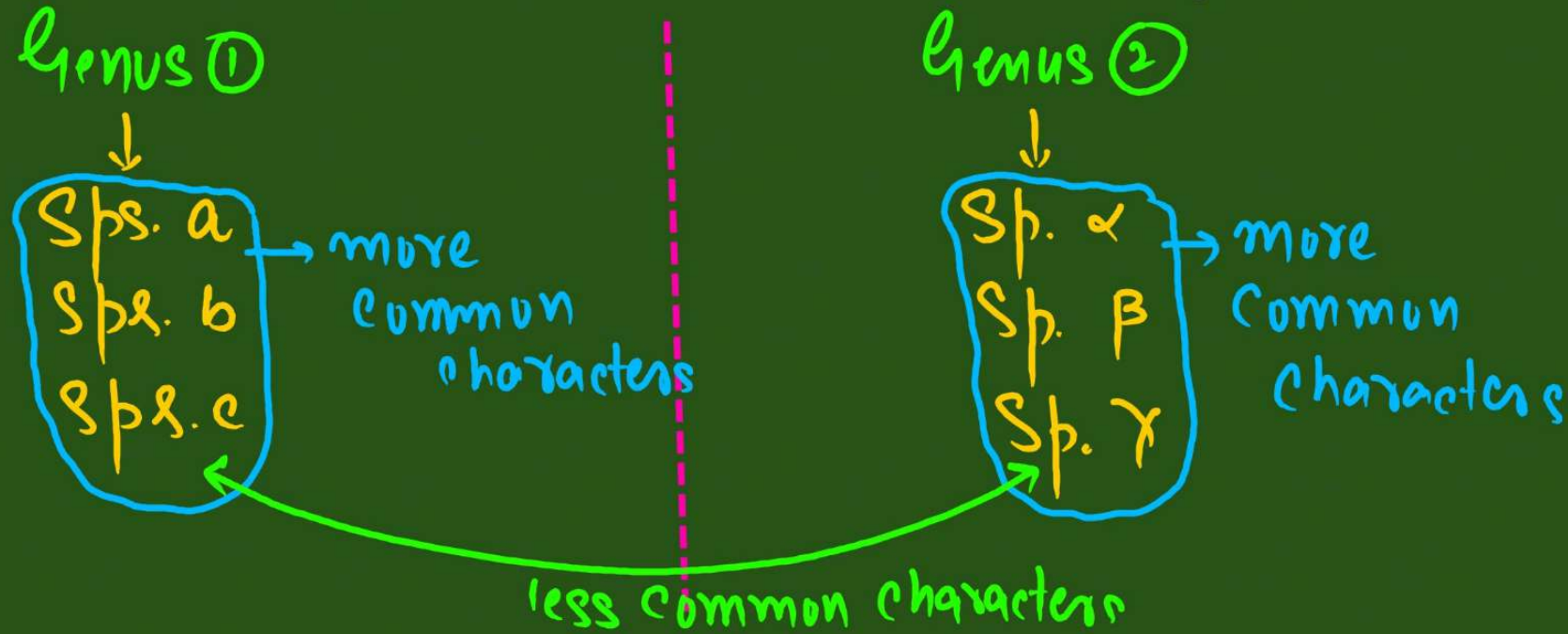
Ernst Mayr
(1904 – 2004)

Born on 5 July 1904, in Kempten, Germany, ERNST MAYR, the Harvard University evolutionary biologist who has been called 'The Darwin of the 20th century', was one of the 100 greatest scientists of all time. Mayr joined Harvard's Faculty of Arts and Sciences in 1953 and retired in 1975, assuming the title *Alexander Agassiz Professor of Zoology Emeritus*. Throughout his nearly 80-year career, his research spanned ornithology, taxonomy, zoogeography, evolution, systematics, and the history and philosophy of biology. He almost single-handedly made the origin of species diversity the central question of evolutionary biology that it is today. He also pioneered the currently accepted definition of a biological species. Mayr was awarded the three prizes widely regarded as the triple crown of biology: the Balzan Prize in 1983, the International Prize for Biology in 1994, and the Crafoord Prize in 1999. Mayr died at the age of 100 in the year 2004.

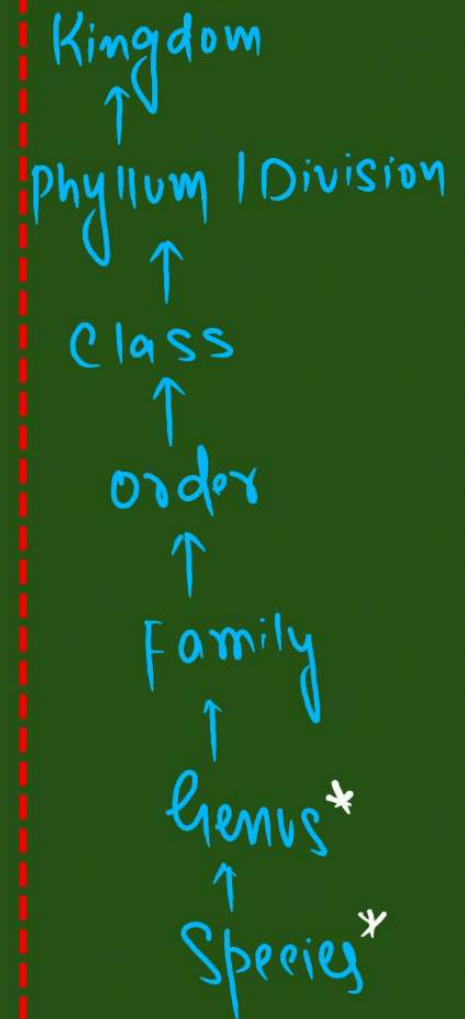


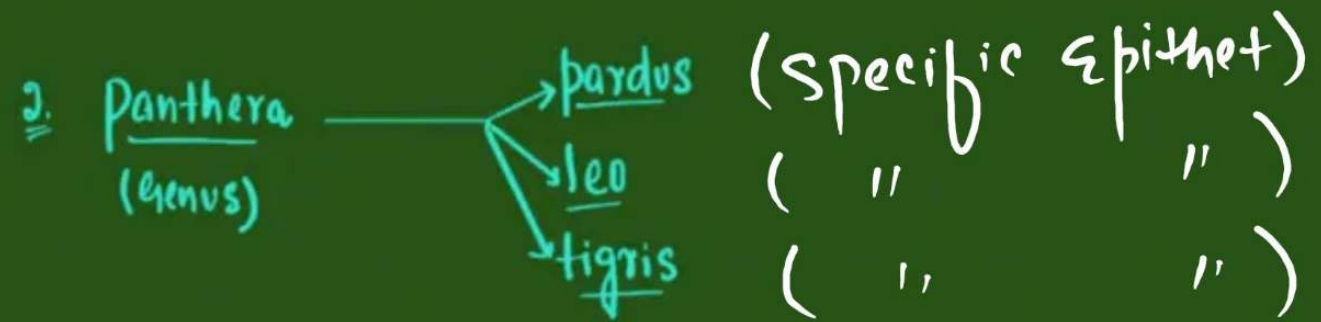
GENUS

- Group of related Species.
- imp Genus has more characters in common in comparison to species of other genera.



Each genus — may have one/more than one specific epithet.





3. Felis — include cats
(genus)

Ex. Felis catus
↓ ↓
Genus Specific epithet.

1.2.2 Genus

Genus comprises a group of related species which has more characters in common in comparison to species of other genera. We can say that genera are aggregates of closely related species. For example, potato and brinjal are two different species but both belong to the genus *Solanum*. Lion (*Panthera leo*), leopard (*P. pardus*) and tiger (*P. tigris*) with several common features, are all species of the genus *Panthera*. This genus differs from another genus *Felis* which includes cats.



FAMILY



≡ group of related genera.

imp ≡ Plant families are characterised on the basis of

a. Vegetative characters
AND
b. Reproductive characters.

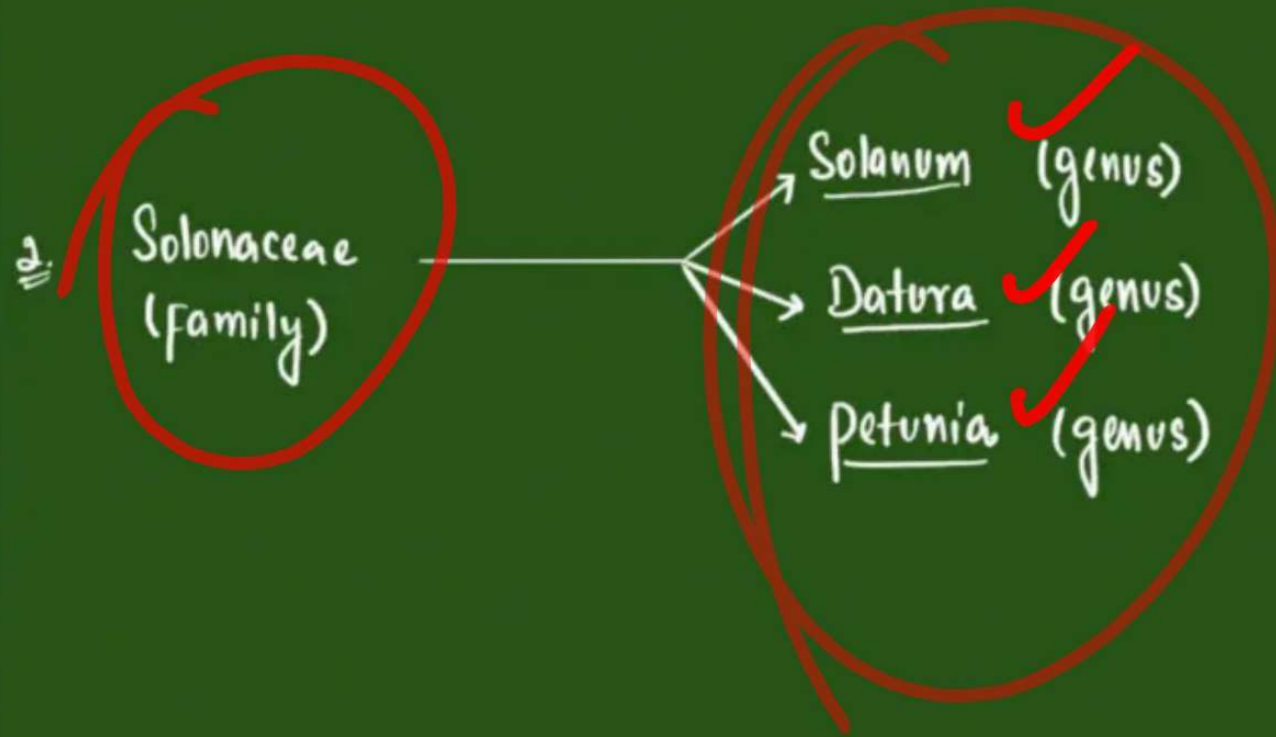
≡ In animal families, suffix -idae is used mostly

ex. Felidae (Cat Family)
Canidae (Dog Family)
Hominidae (Human Family)
Muscidae (Housefly Family)

≡ In plant families, suffix -aceae is used mostly.

ex. Solanaceae (Potato Family) ✓
Liliaceae (Lily Family) ✓
Fabaceae (Bean Family) ✓
Poaceae (Grass Family) ✓

Examples



1.2.3 Family

The next category, **Family**, has a group of related genera with still less number of similarities as compared to genus and species. Families are characterised on the basis of both vegetative and reproductive features of plant species. Among plants for example, three different genera *Solanum*, *Petunia* and *Datura* are placed in the family Solanaceae. Among animals for example, genus *Panthera*, comprising lion, tiger, leopard is put along with genus, *Felis* (cats) in the family Felidae. Similarly, if you observe the features of a cat and a dog, you will find some similarities and some differences as well. They are separated into two different families – Felidae and Canidae, respectively.

ORDER



≡ group of related Families.

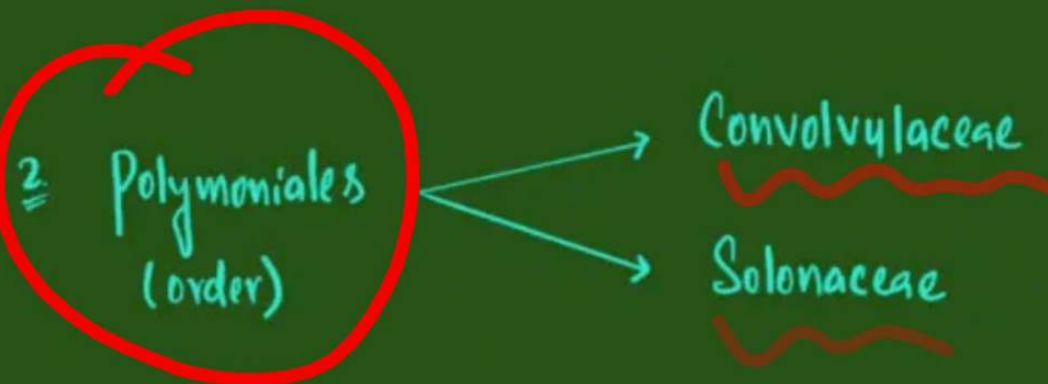
≡ Some animal orders are: Primata, Diptera, Rodentia, Carnivora.

≡ In plant orders, suffix "-ales" is used

Ex. Sapindales, Poales, Polymoniales.

imp ≡ Plant orders — are formed on the basis of floral characters → Floral characters.

EXAMPLES:



NCERT READING

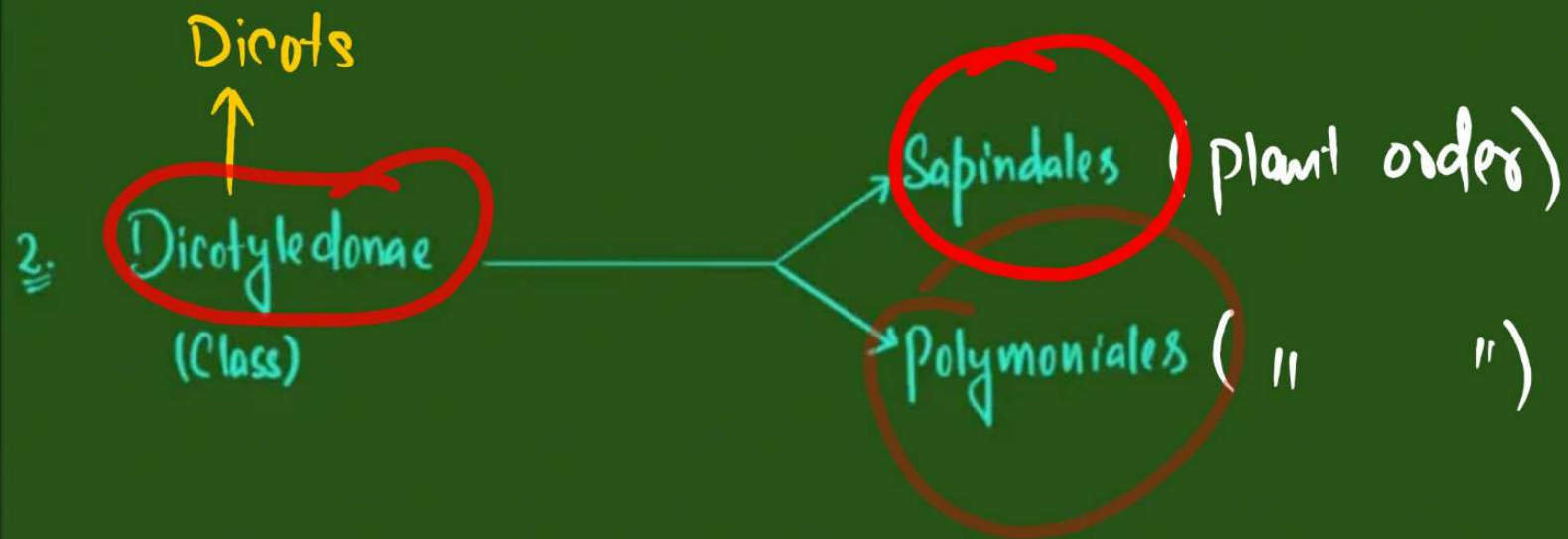
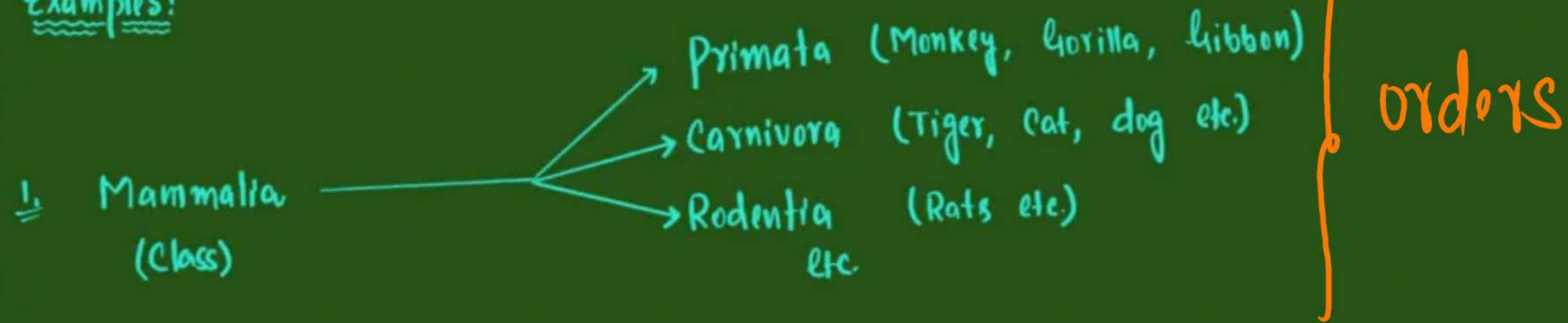
1.2.4 Order

You have seen earlier that categories like species, genus and families are based on a number of similar characters. Generally, order and other higher taxonomic categories are identified based on the aggregates of characters. Order being a higher category, is the assemblage of families which exhibit a few similar characters. The similar characters are less in number as compared to different genera included in a family. Plant families like Convolvulaceae, Solanaceae are included in the order Polymoniales mainly based on the floral characters. The animal order, Carnivora, includes families like Felidae and Canidae.

CLASS

≡ group of related orders.

Examples:



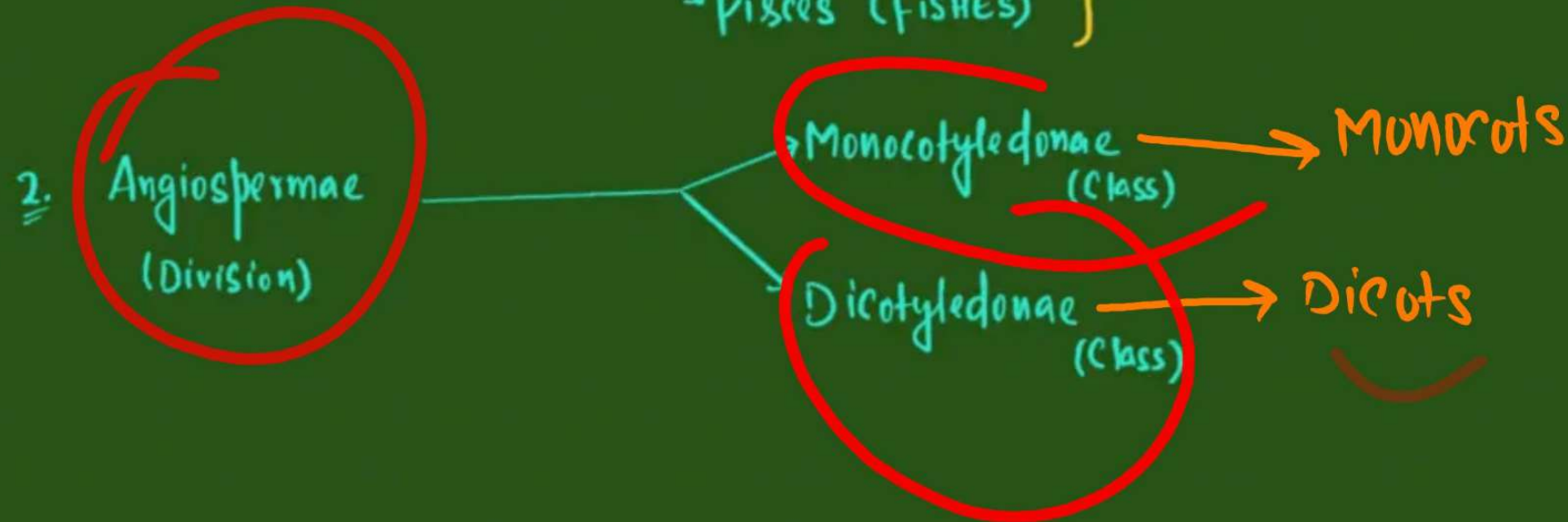
1.2.5 Class

This category includes related orders. For example, order Primata comprising monkey, gorilla and gibbon is placed in class Mammalia along with order Carnivora that includes animals like tiger, cat and dog. Class Mammalia has other orders also.

PHYLLUM/DIVISION

- ≡ Phylum — used in animal classification.
- Division — used in plant classification.
- ≡ Phylum / Division — group of related classes.

Examples:



1.2.6 Phylum

Classes comprising animals like fishes, amphibians, reptiles, birds along with mammals constitute the next higher category called Phylum. All these, based on the common features like presence of notochord and dorsal hollow neural system, are included in phylum Chordata. In case of plants, classes with a few similar characters are assigned to a higher category called Division.

KINGDOM

≡ group of related Divisions / Phyla.
≡ highest taxonomic Category.

Example :

1. Plant Kingdom — includes All plants.
2. Animal Kingdom — includes All Animals.

Plant Kingdom = group of related
Divisions.

Animal Kingdom = group of related
Phyla.

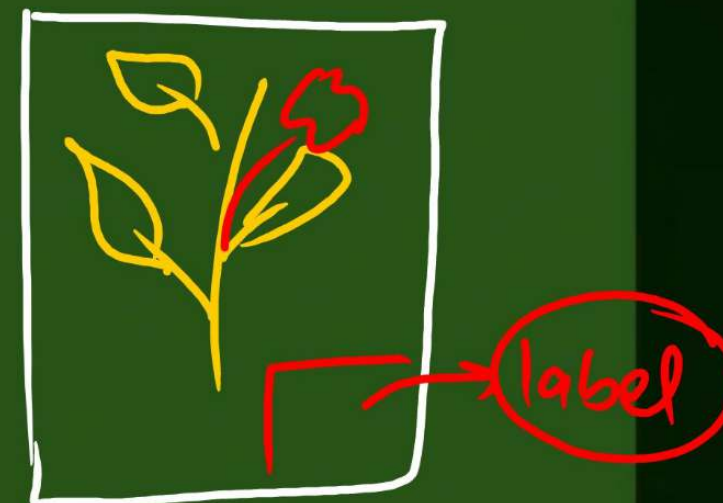
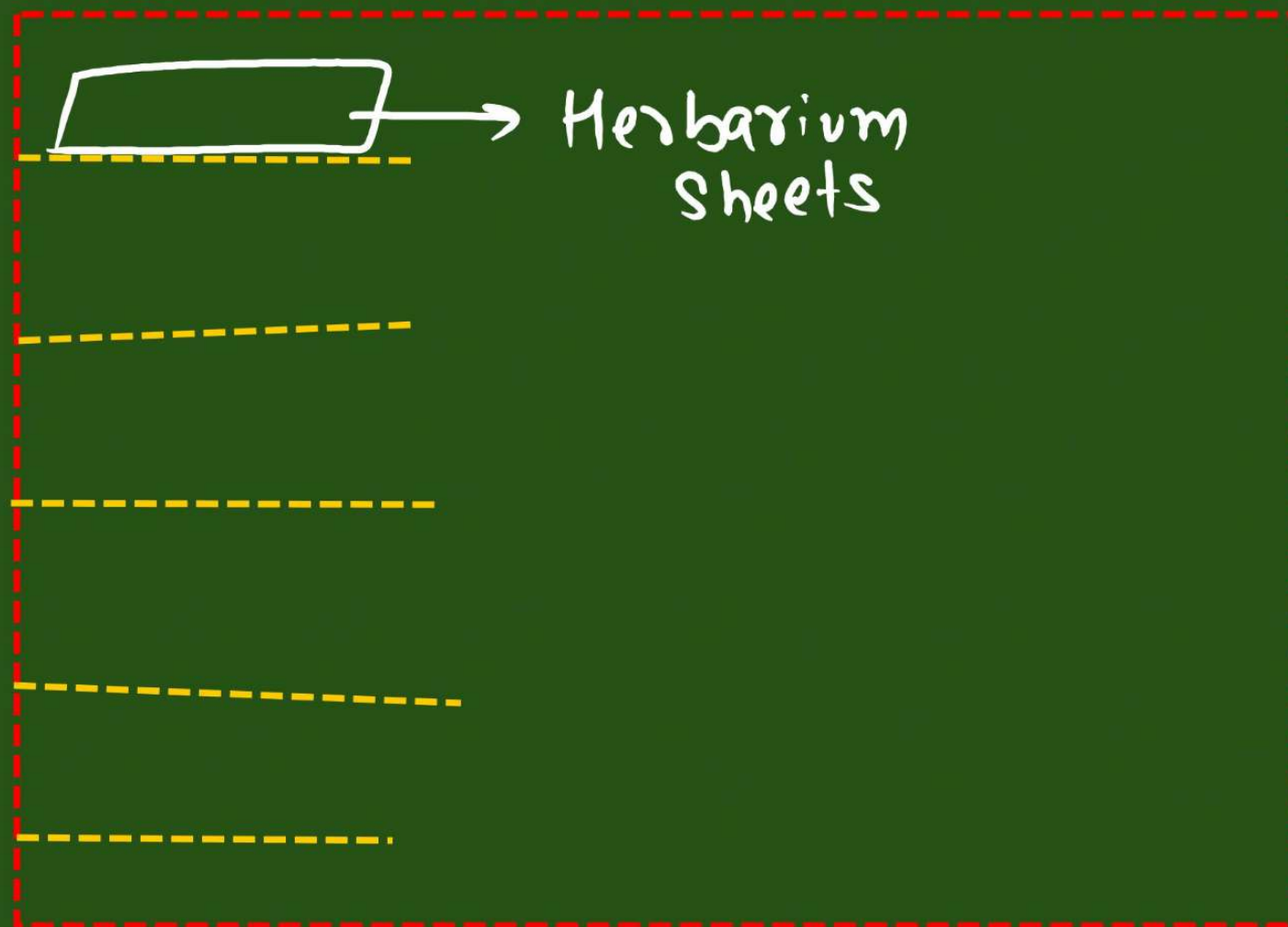
NCERT READING

1.2.7 Kingdom

All animals belonging to various phyla are assigned to the highest category called Kingdom Animalia in the classification system of animals. The Kingdom Plantae, on the other hand, is distinct, and comprises all plants from various divisions. Henceforth, we will refer to these two groups as animal and plant kingdoms.

TABLE 1.1 Organisms with their Taxonomic Categories

Common Name	Biological Name	Genus	Family	Order	Class	Phylum/ Division
Man	<i>Homo sapiens</i>	<i>Homo</i>	Hominidae	Primata	Mammalia	Chordata
Housefly	<i>Musca domestica</i>	<i>Musca</i>	Muscidae	Diptera	Insecta	Arthropoda
Mango	<i>Mangifera indica</i>	<i>Mangifera</i>	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	<i>Triticum aestivum</i>	<i>Triticum</i>	Poaceae	Poales	Monocotyledonae	Angiospermae



HERBARIUM

Herbarium sheet.



label

Herbarium



≡ It is a store house of "dead plant specimens only".

(Specimen = Sample)

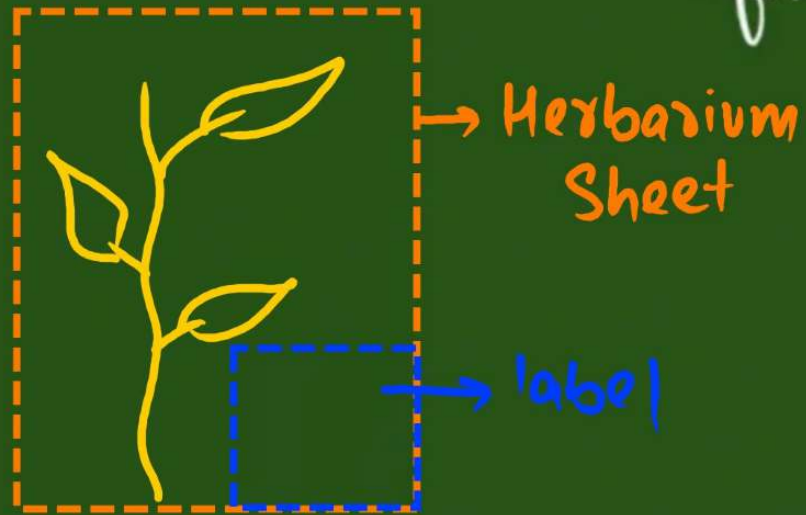
≡ In Herbarium, plant specimens are —

- Collected
- Dried
- pressed
- and preserved on sheets.

≡ Herbarium — act as a quick referral system for classification.

≡ Herbarium Sheets — are arranged according to universally accepted system of classification.

≡ Herbarium Sheets — Carry a label (on right hand bottom) providing information about



- Date and place of Collection. ✓
- English name, Common name, Botanical name.
- Family ✓
- Collector's name. ✓

≡ International Size of herbarium Sheet = 29 x 41.5 cm.



QUESTIONS

Basic unit or smallest taxon of taxonomy/classification is:

[1990]

- ☒ 1 Species
- ☐ 2 Kingdom
- ☐ 3 Family
- ☐ 4 Variety

QUESTIONS

A taxon is:

[1992]

- 1 A group of related families ✓
- 2 A group of related species ✓
- 3 A type of living organisms ✓
- 4 A taxonomic group of any ranking ✓

QUESTIONS

Species are considered as:

[OS 2003]

- ☒ 1 Real basic units of classification
- ☐ 2 The lowest units of classification
- ☐ 3 Artificial concept of human mind
- ☐ 4 Real units of classification devised by taxonomists

QUESTIONS

H/w

House fly belongs to _____ family.

[2023 Manipur]

- 1 Cyprinidae
- 2 Hominidae
- 3 Calliphoridae
- 4 Muscidae

QUESTIONS

H/w.

Which one of the following animals is correctly matched with its particular named taxonomic category?
[MR* 2011 Pre]

- 1 Housefly – *Musca*, an order
- 2 Tiger – *Tigris*, the species
- 3 Cuttlefish – Mollusca, a class
- 4 Humans – Primata, the family

QUESTIONS

H/w

Match Column-I with Column-II for housefly classification and select the correct option using the codes given below: [MR* 2016 – II]

- 1** A-iv, B-iii, C-ii, D-i
- 2** A-iv, B-ii, C-i, D-iii
- 3** A-iii, B-i, C-iv, D-ii
- 4** A-iii, B-ii, C-iv, D-i

Column-I		Column-II	
A.	Family	(i)	Diptera
B.	Order	(ii)	Arthropoda
C.	Class	(iii)	Muscidae
D.	Phylum	(iv)	Insecta

HOMEWORK

3 Questions



Thank
You